AN306 Group 7

NLP Analysis: Enhancing the Job Search Process

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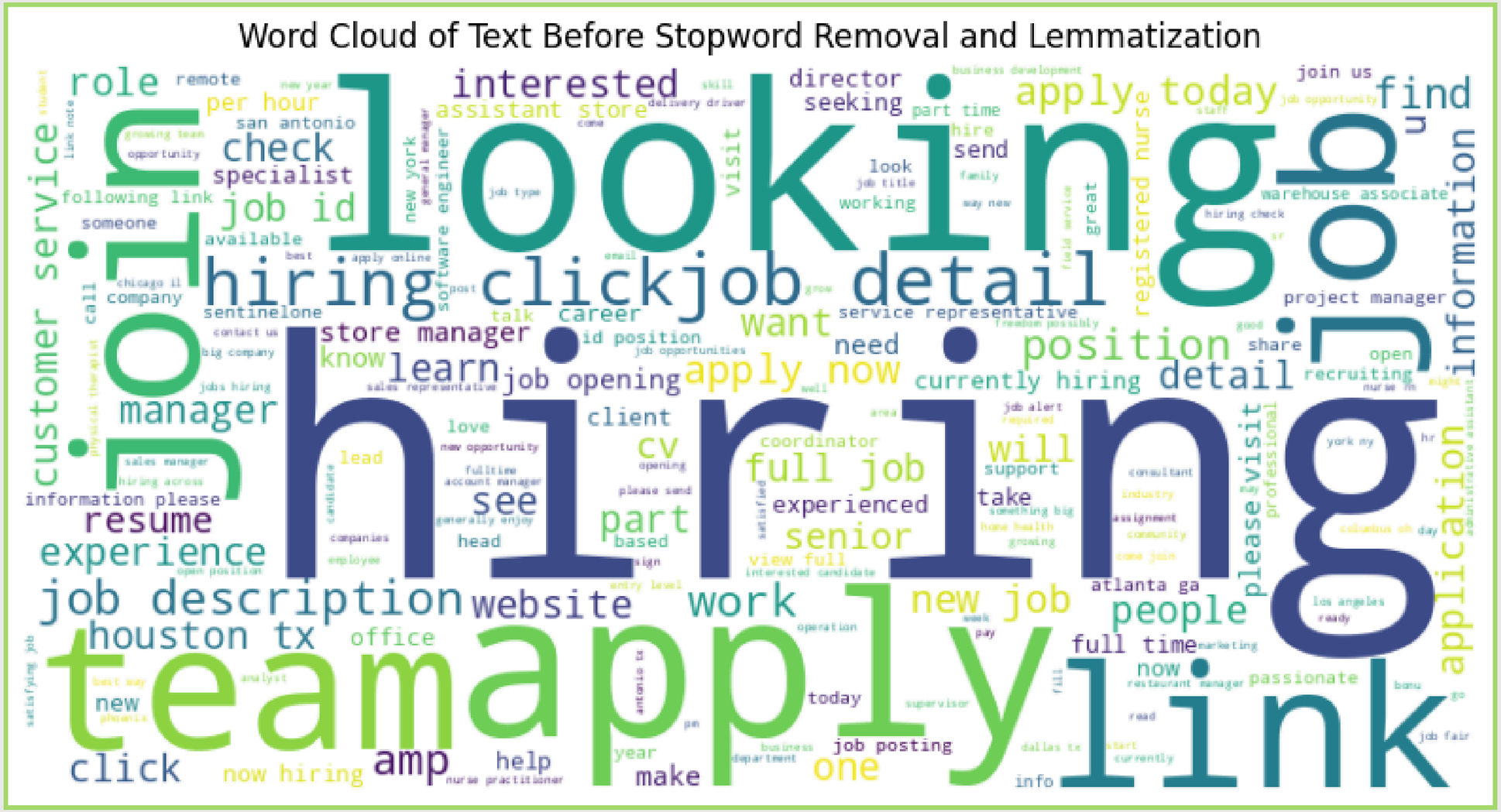
What specific roles and locations are most common among finance and accounting job postings in major U.S. financial hubs on X?

Before preprocessing...



Preprocessing

1. Text cleaning, tokenization, and lemmetization
2. Removal of basic and custom stop words
3. Keyword search and regex for finance-related keywords
4. Location data extraction from clean text

   Apply keyword search and regex for location keywords
5. TF-IDF vectorization, cosine similarity, and deduplication (threshold of ≥ 0.95)

After preprocessing...

815 Tweets

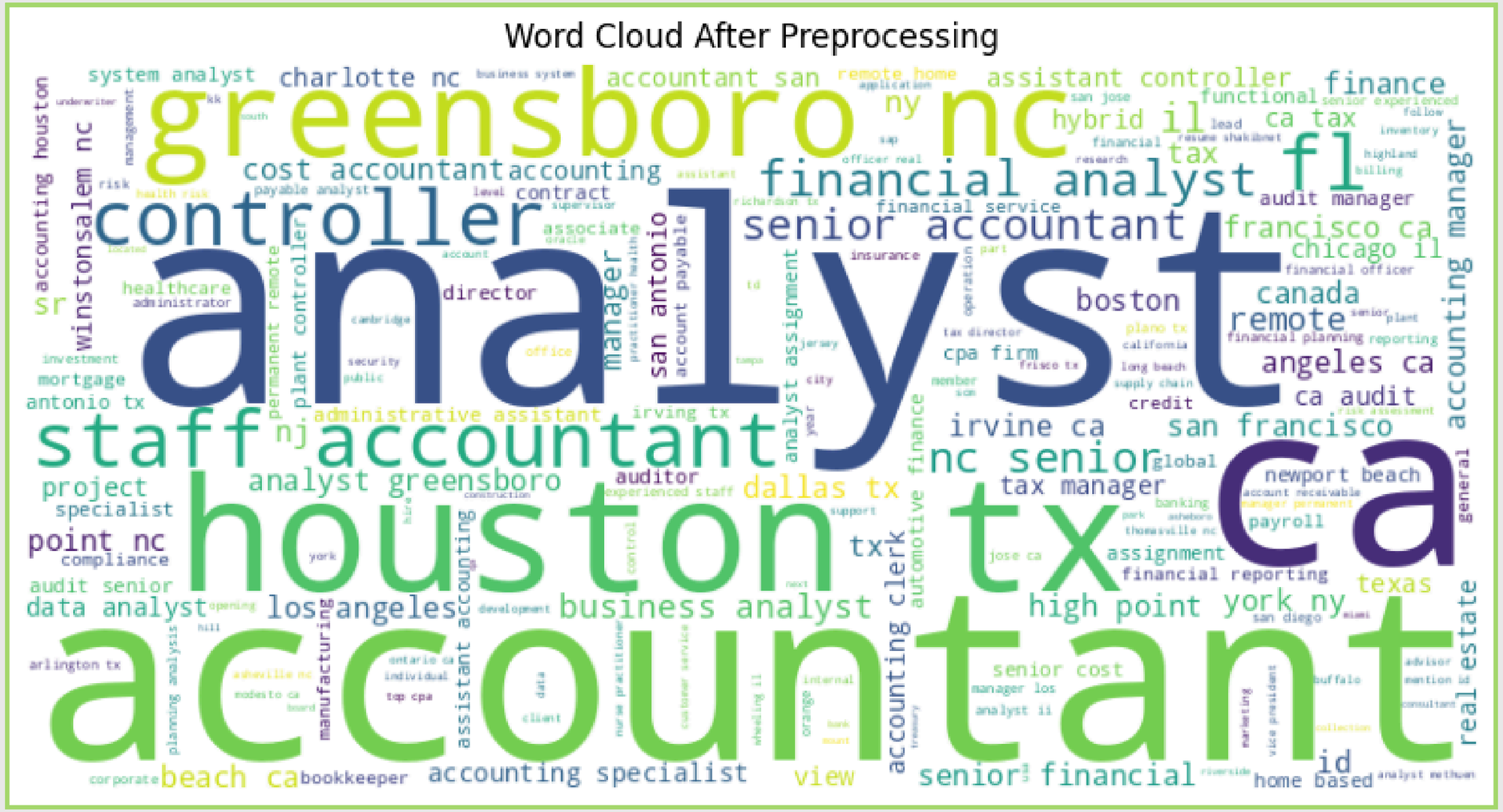
→ 604 tweets after deduplication

Keywords:

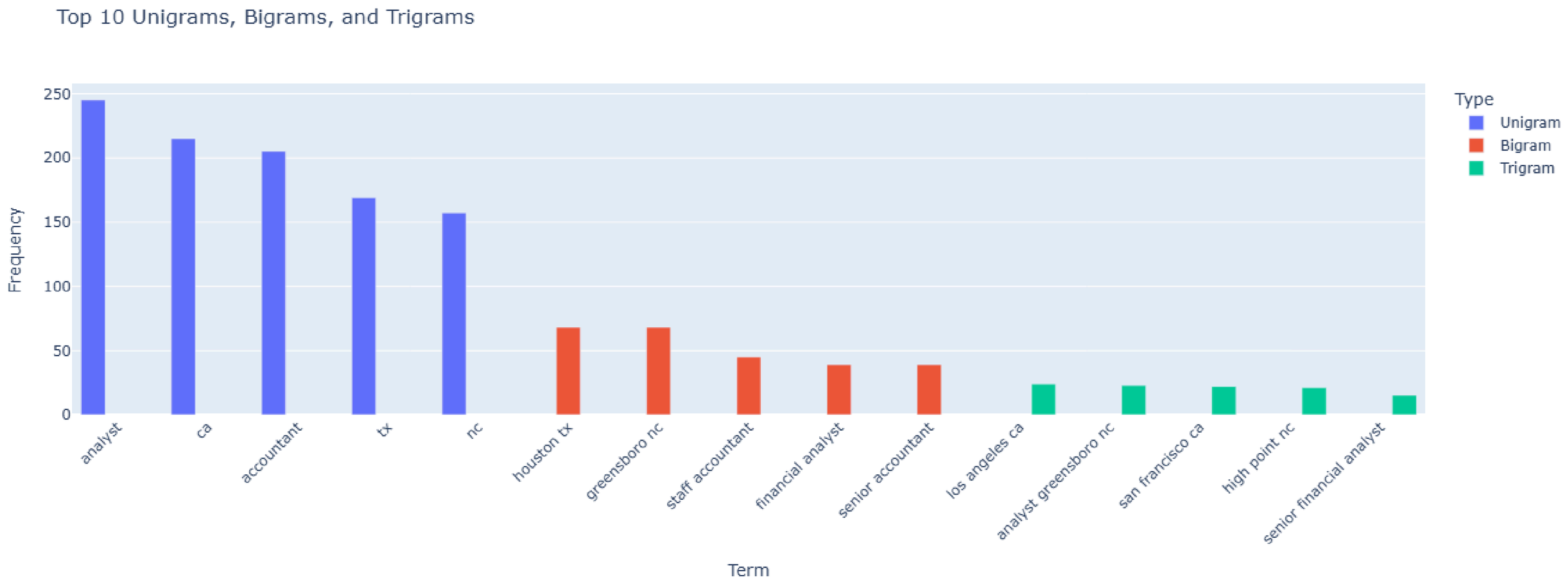
Finance: 'finance', 'financial', 'accounting', 'accountant', 'analyst', 'financial analyst', 'banking', 'bank', 'investment', 'investment analyst', 'asset management', 'portfolio', 'cpa', 'audit', 'auditor', 'tax', 'treasury', 'risk', 'credit', 'loan officer', 'underwriter', 'controller', 'bookkeeper'

Locations: 'boston', 'chicago', 'new york', 'nyc', 'los angeles', 'san francisco', 'san diego', 'houston', 'dallas', 'austin', 'ma', 'massachusetts', 'il', 'illinois', 'ny', 'new york', 'ca', 'california', 'tx', 'texas', 'fl', 'florida', 'nc', 'north carolina', 'nj', 'new jersey', ‘miami’, ‘charlotte’, ‘newark’, ‘jersey city’

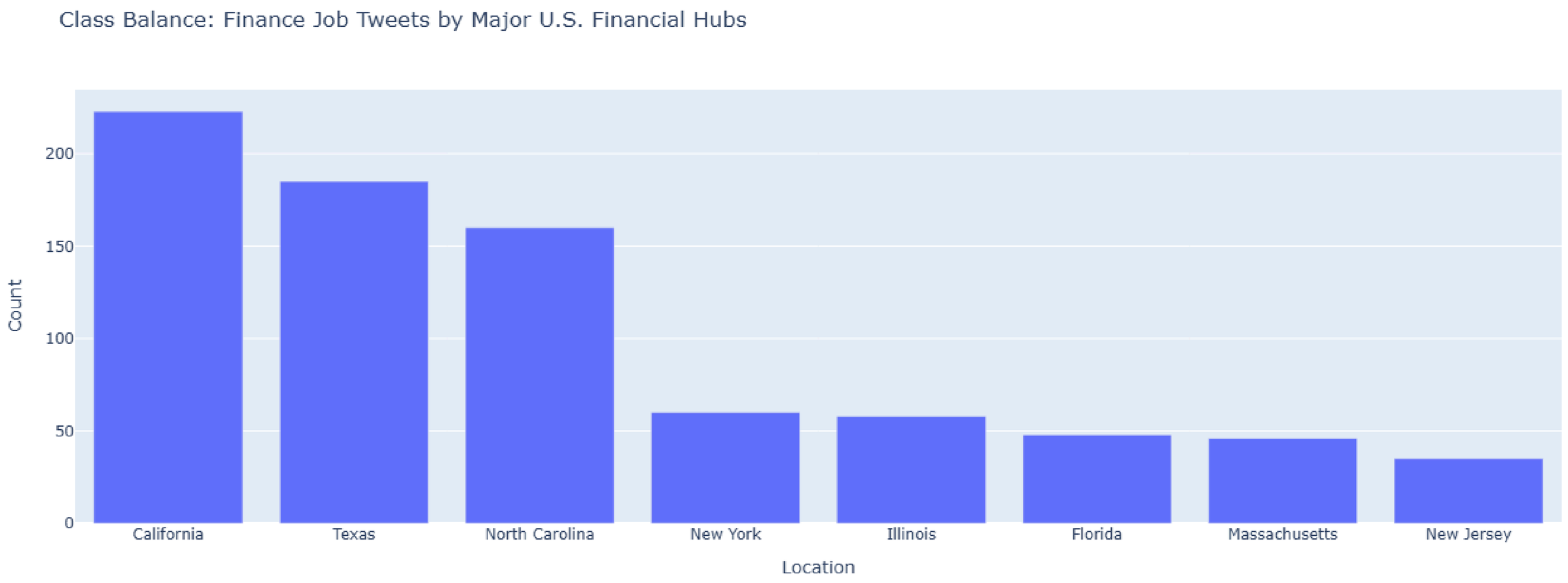
After preprocessing...


Word Cloud After Preprocessing
analyst
accountant
greensboro nc
controller
staff accountant
houston tx
ca
fl
financial analyst
senior accountant
system analyst
charlotte nc
accountant san
assistant controller
finance
hybrid
il
ca tax
tax
cost accountant
accounting
francisco ca
chicago il
boston
canada
remote
angeles ca
ca audit
irvine ca
san francisco
nc senior
tax manager
newport beach
payroll
assignment
financial reporting
texas
high point
york ny
business analyst
los angeles
analyst greensboro
dallas tx
tx
project specialist
point nc
compliance
audit senior
data analyst
beach ca
bookkeeper
accounting specialist
view
senior financial
home based
id
real estate
general
accounting manager
winstonsalem nc
accounting houston
sr
healthcare
mortgage
antonio tx
director
manager
san antonio
assistant accounting
manufacturing
plant controller
permanent remote
administrative assistant
nj
irving tx
auditor
accounting clerk
automotive finance
analyst assignment
senior cost
audit manager
functional
cpa firm
credit
global
banking
jose ca
arlington tx
planning analysis
modesto ca
top cpa
individual
ontario ca
corporate
nurse practitioner
customer service
wheeling il
internal
orange
client
data
asheville nc
board
opening
hill
york
control
next
development
construction
experienced staff
support
hire
park
thomasville nc
manager permanent
account receivable
riverside
buffalo
miami
san diego
marketing
vice president
advisor
mention id
consultant
collection
analyst nathan
inventory
highland
billing
senior experienced
follow
lead
resume shakibnet
financial
research
officer real
assistant
richardson tx
supervisor
level
payable analyst
contract
risk
health risk
management
undewriter
kk
south
business system
application
remote home
ny
sap
financial service
insurance
part
operation
account payable
practitioner health
id
cambridge
security
public
office
jersey
city
year
tampa
member
son
tax director
plano tx
friscо tx
long beach
supply chain
risk assessment
located
account
oracle
investment
administrator
plant
senior
reporting
financial planning
california
financial officer
nc
jose ca
thomasville nc
manager permanent
account receivable
banking
park
asheboro
support
next
development
construction
experienced staff
hire
control
york
opening
hill
board
modesto ca
asheville nc
corporate

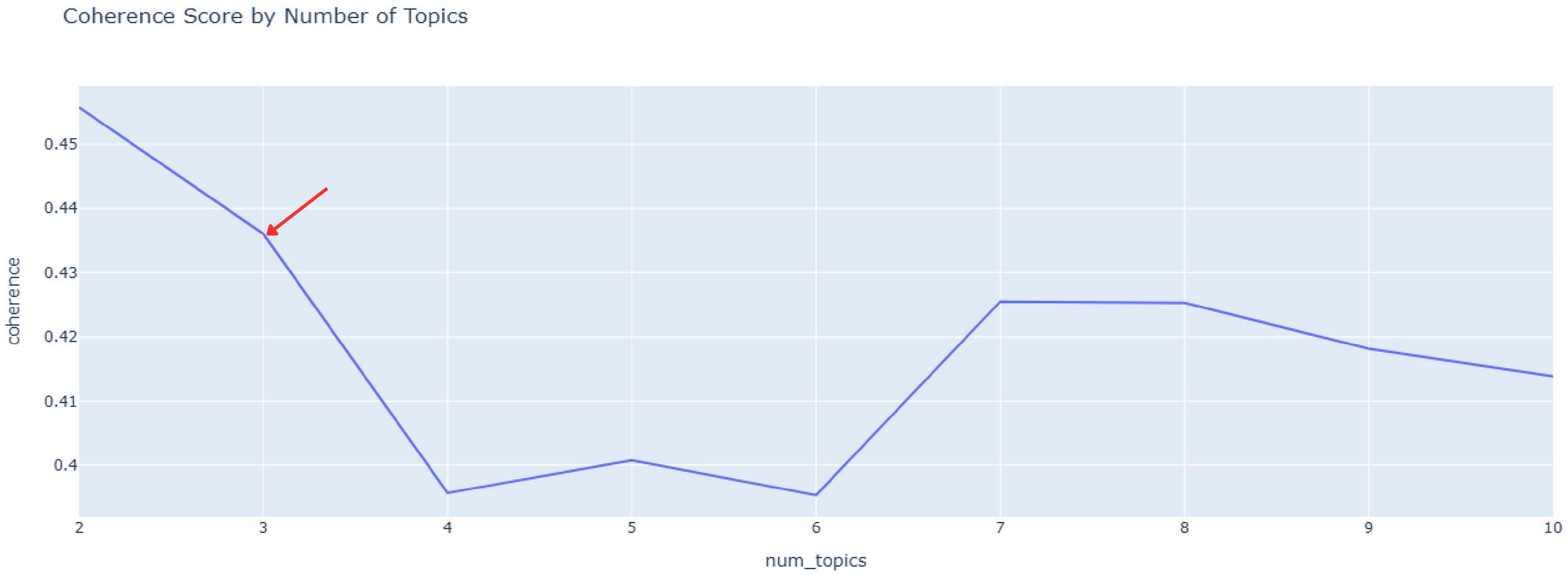
Unigram, Bigram, and Trigram Chart



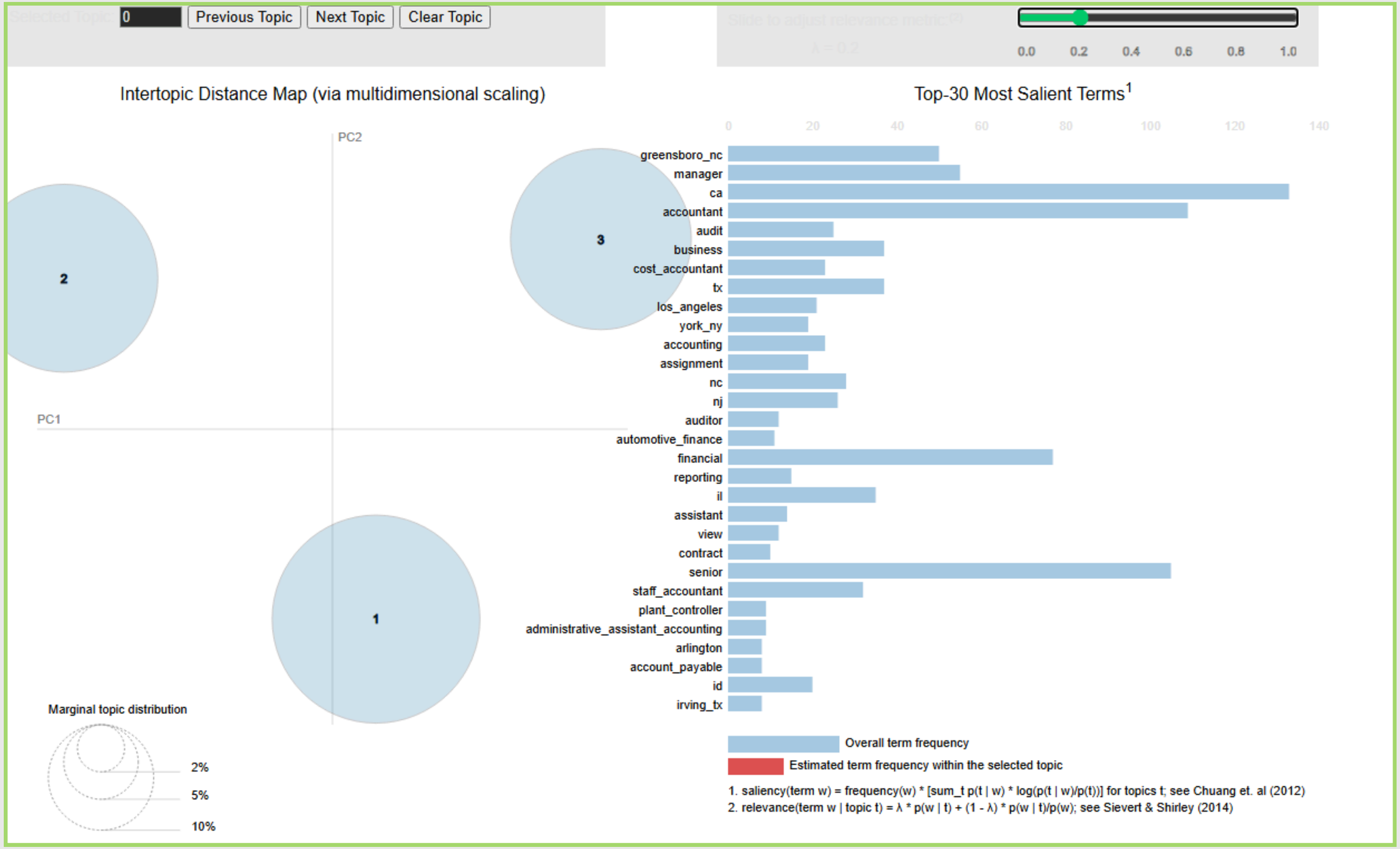
Class Balance Chart



LDA Modeling



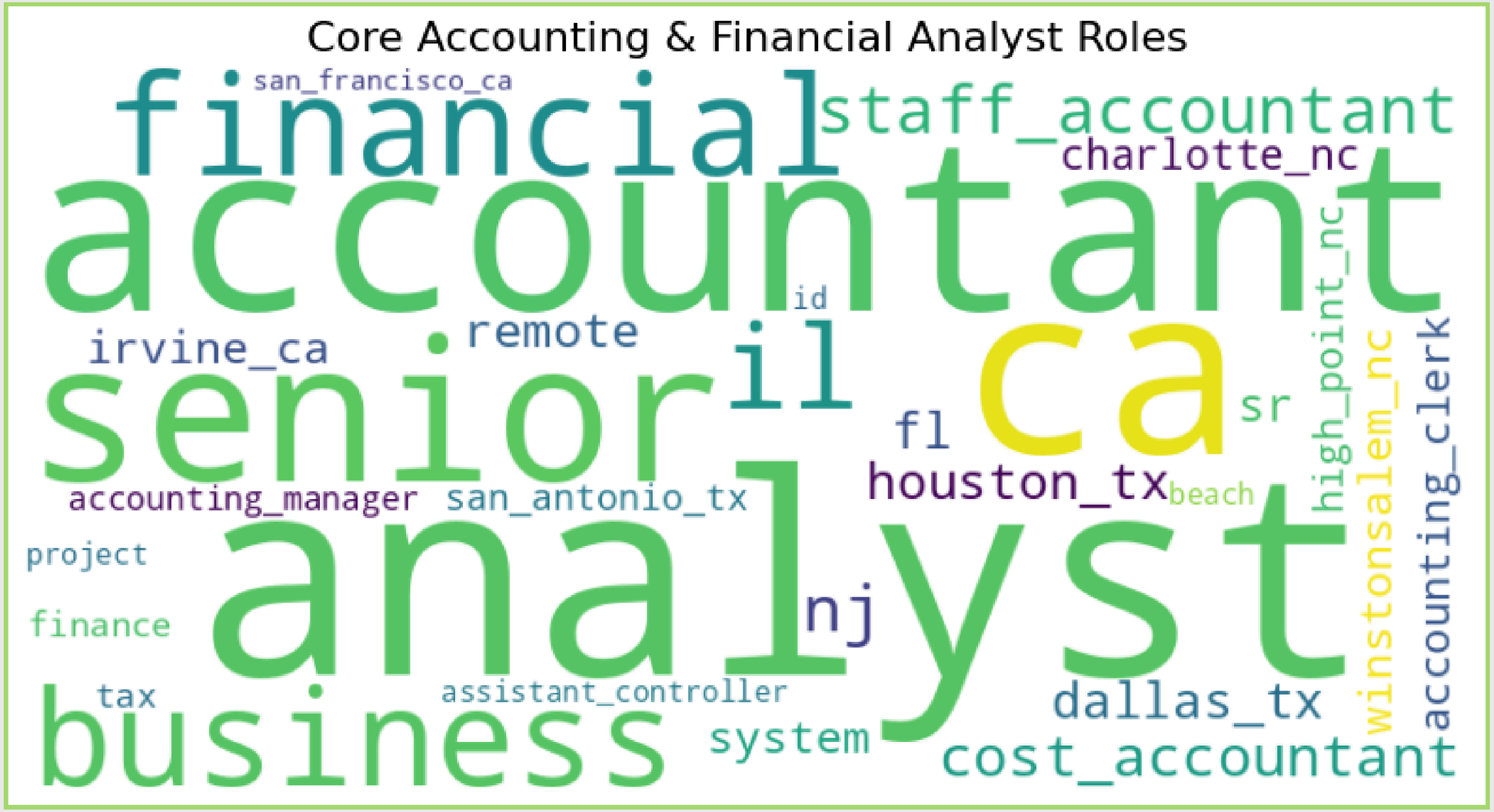
LDA Modeling



Number of topics: 3

λ: 0.20, increases the interpretability of the topics

Topic 1: Core Accounting & Financial Analyst Roles Across U.S. Markets



General/Core Roles: analyst, accountant, staff_accountant, cost_accountant

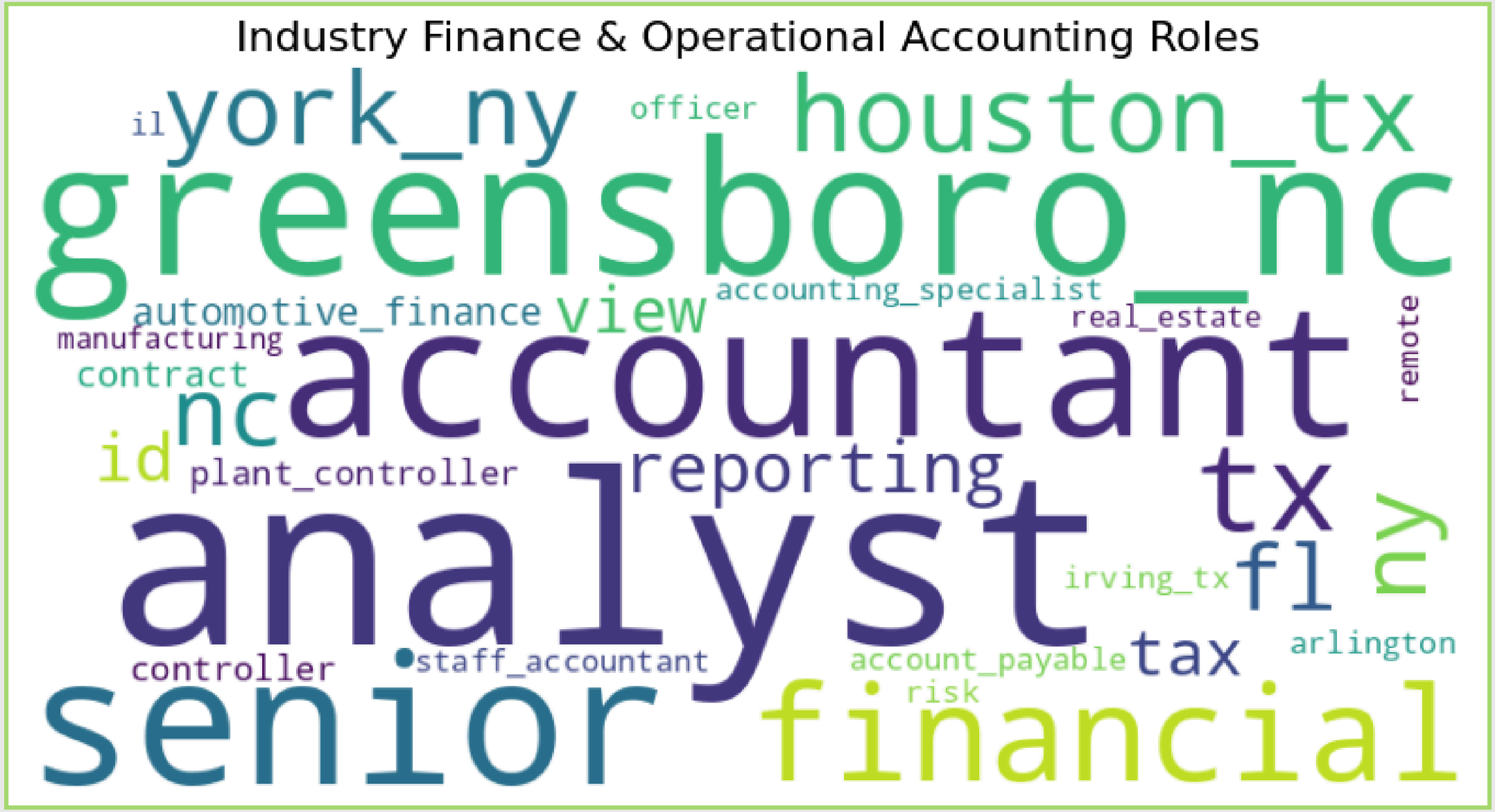
Mixed Locations: charlotte, houston, san antonio (major hubs) and irvine, winston-salem, high point (regional hubs)

Topic 2: Senior Audit, Tax & Remote Roles Concentrated in Major Financial Hubs

Senior/Specialized Roles: manager, audit, tax, controller, senior

Remote Terms: permanent remote home based, remote, hybird

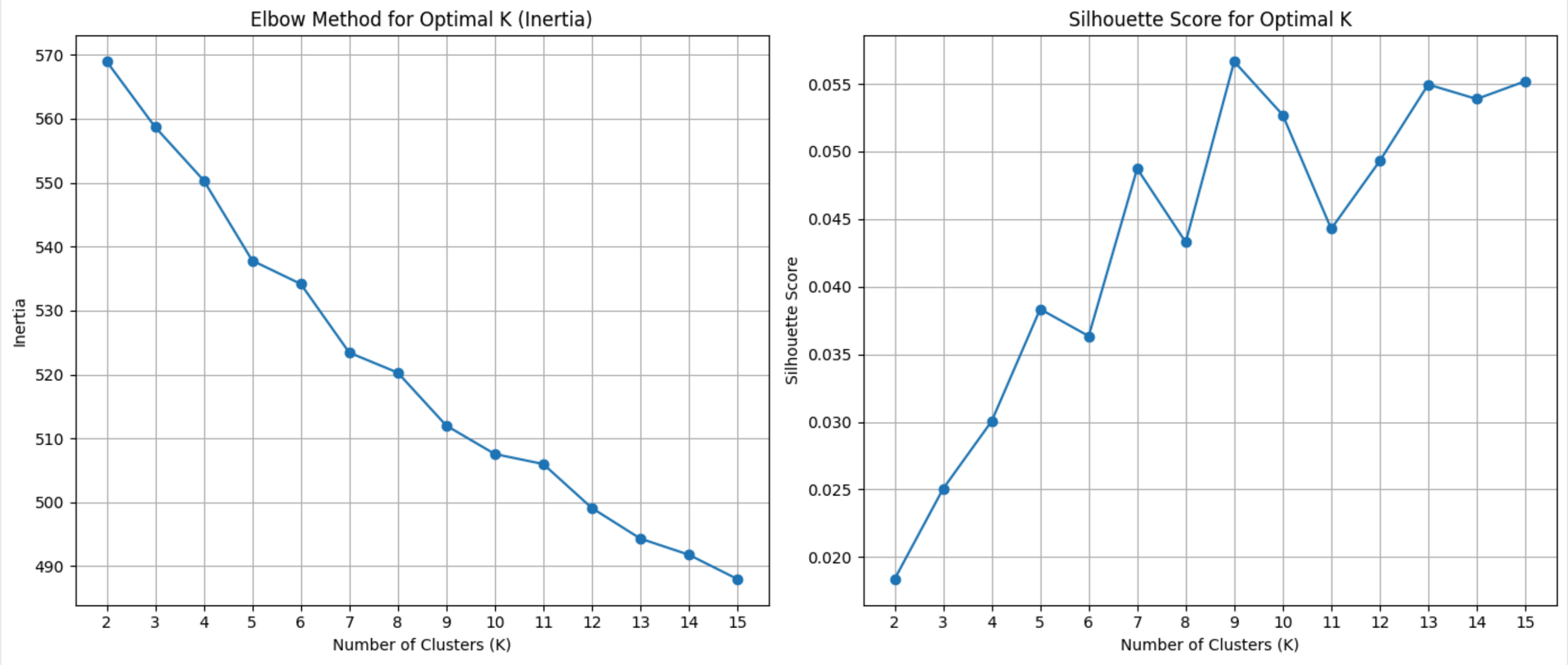
Topic 3: Industry-Specific Finance Roles in Major U.S. Business and Financial Centers



Industry / Operations Roles: plant_controller, manufacturing, AP/AR, real estate

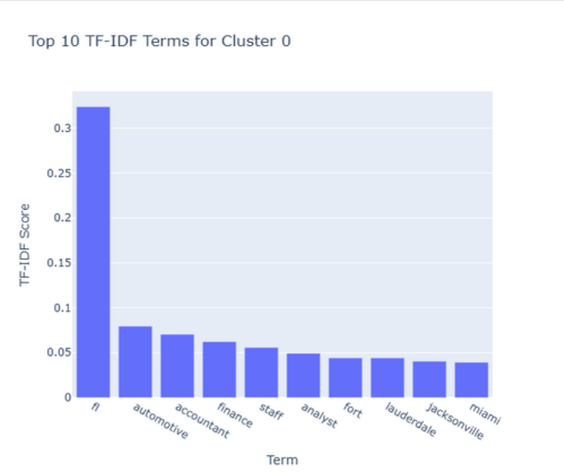
Key Business Centers: new york, houston, greensboro

Silhouette Scoring for K-Means Clustering

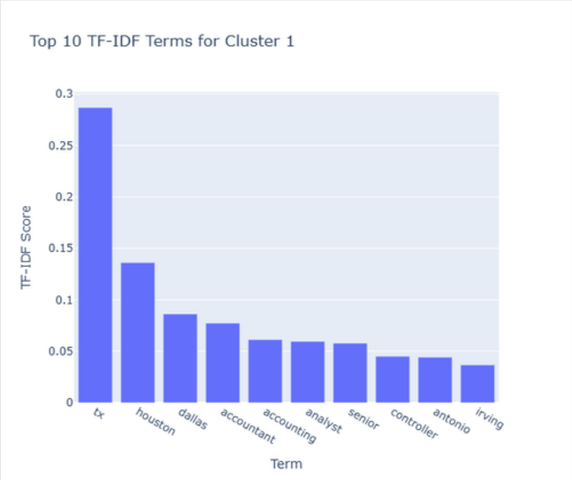


k = 9

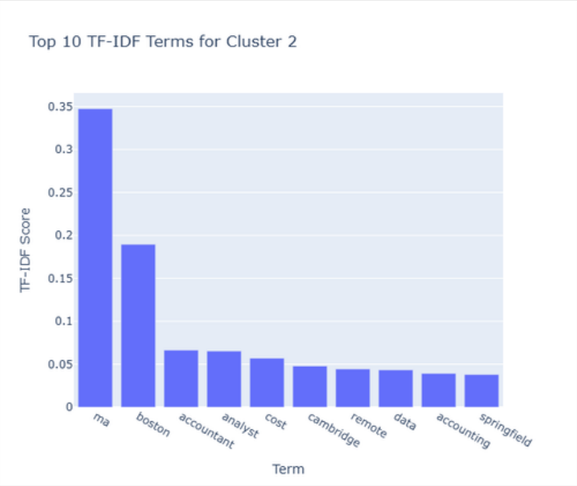
TF-IDF Terms



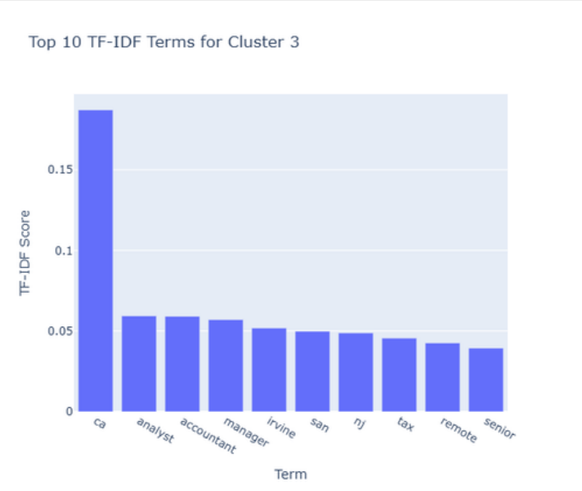
Florida
Staff & Auto-Sector
Accounting



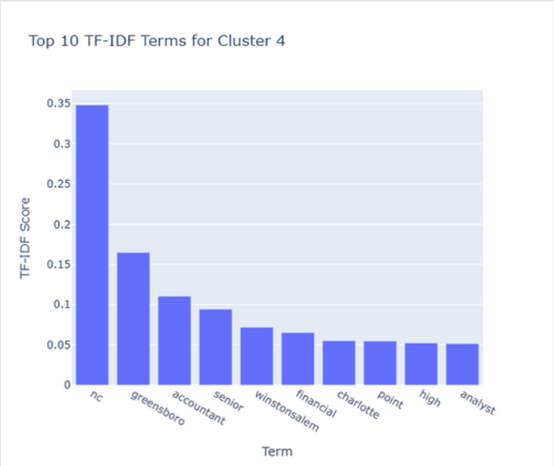
Texas
Senior Controllers
& Analysts



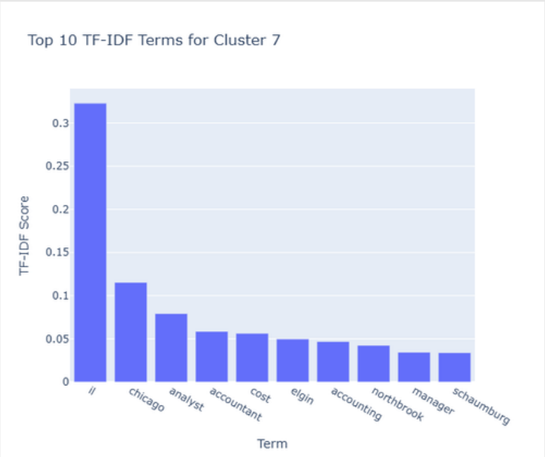
Massachusetts
Cost & Data Accounting



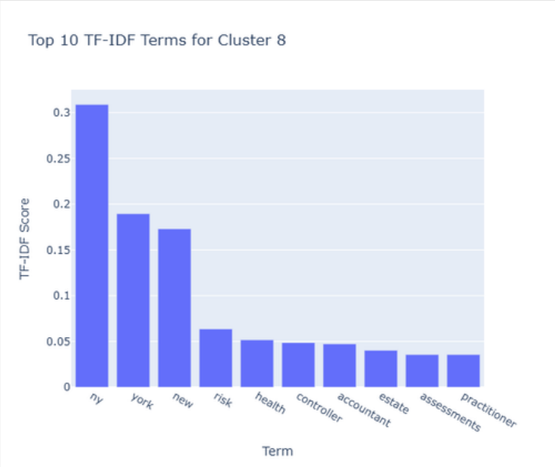
California
Tax & Remote Analyst Roles



North Carolina
Senior Financial Specialists

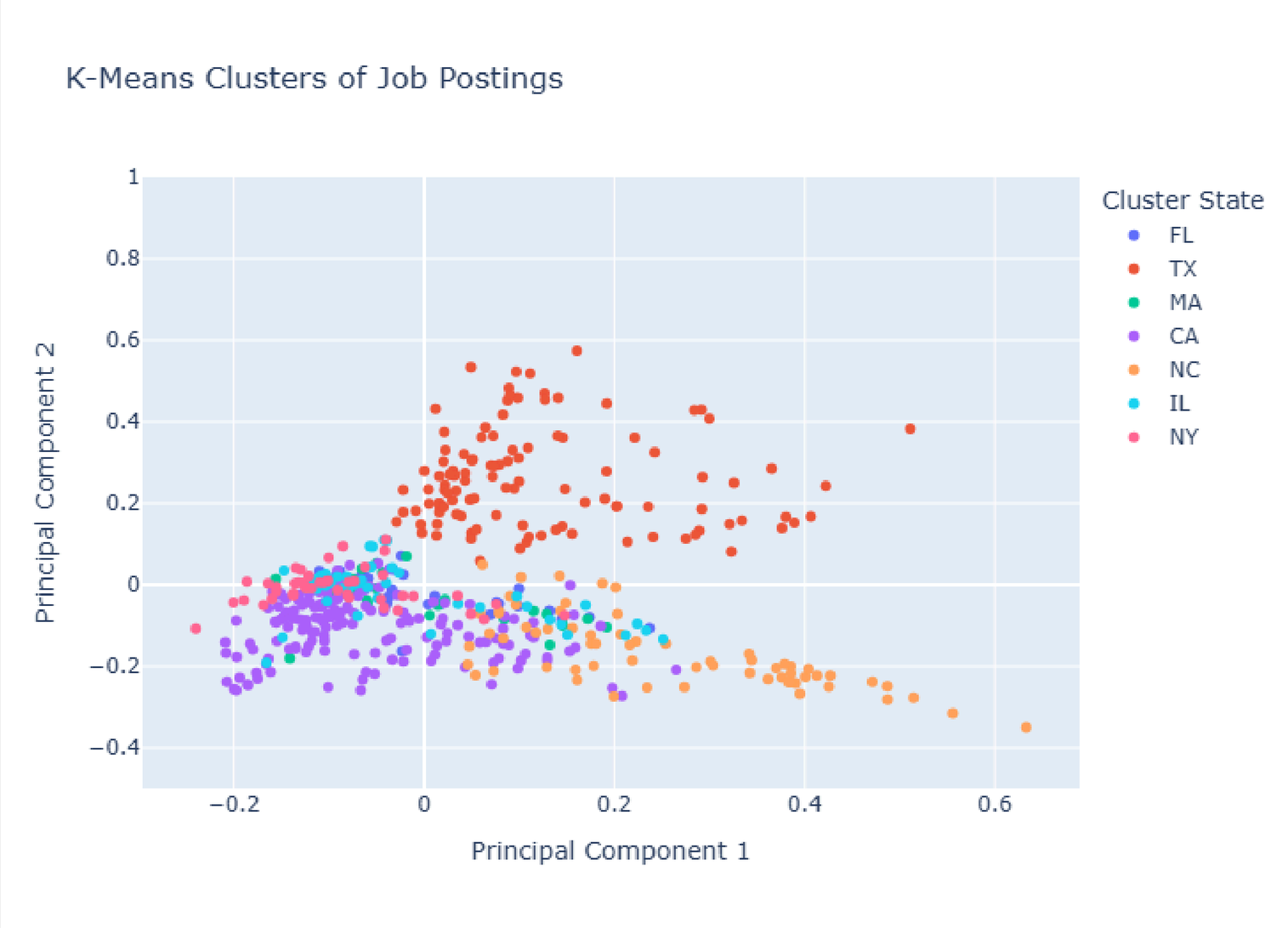


Illinois
Cost Accounting & Management



New York
Real Estate & Healthcare Finance

K-Means Clustering



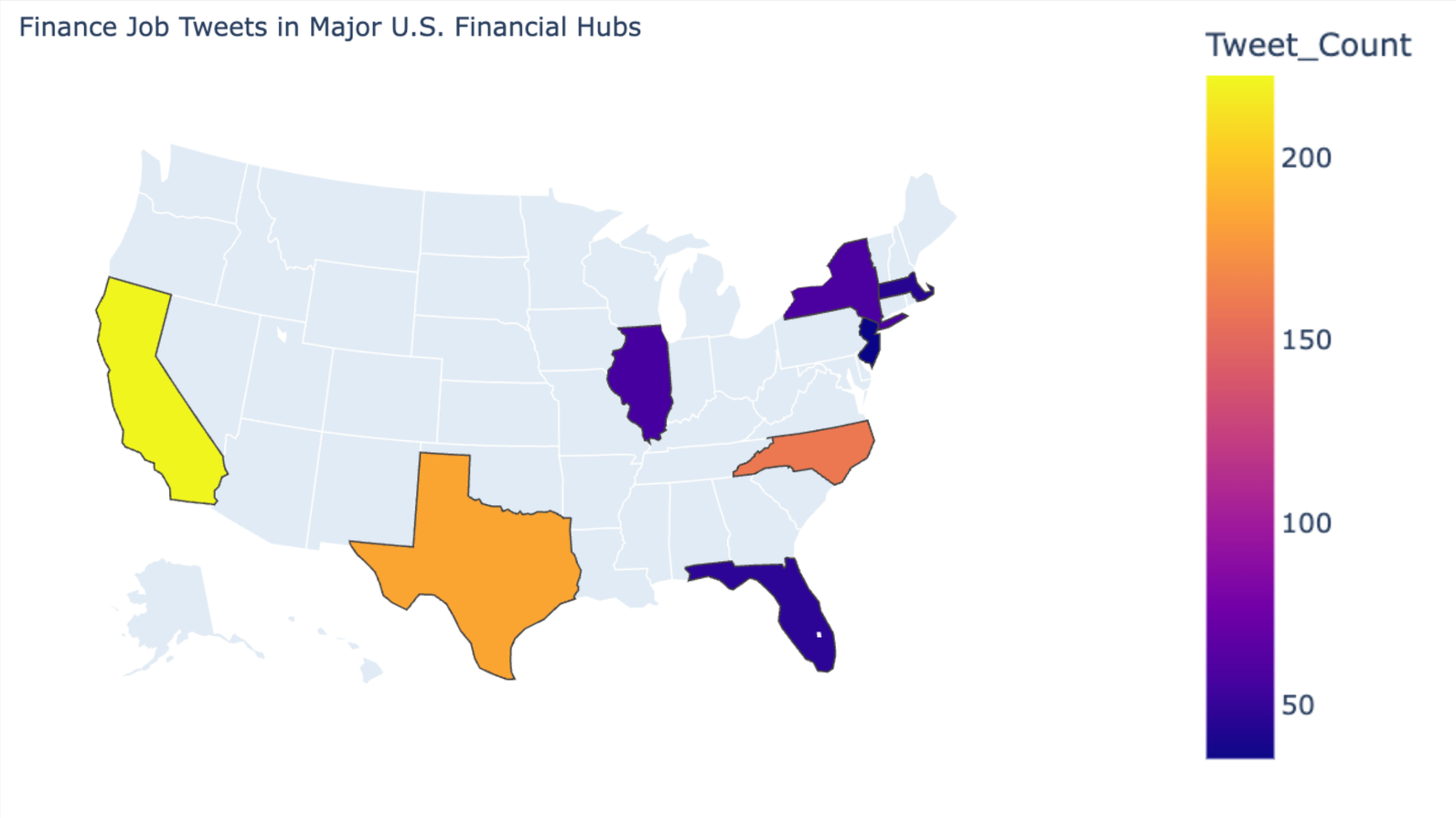
Geospatial Analytics

- Extracted location from both user Location field and tweet text
- Created new variable: state_from_text

- Combined sources into Final_State:
  → Prioritized Location field
  → Filled missing values using text extraction

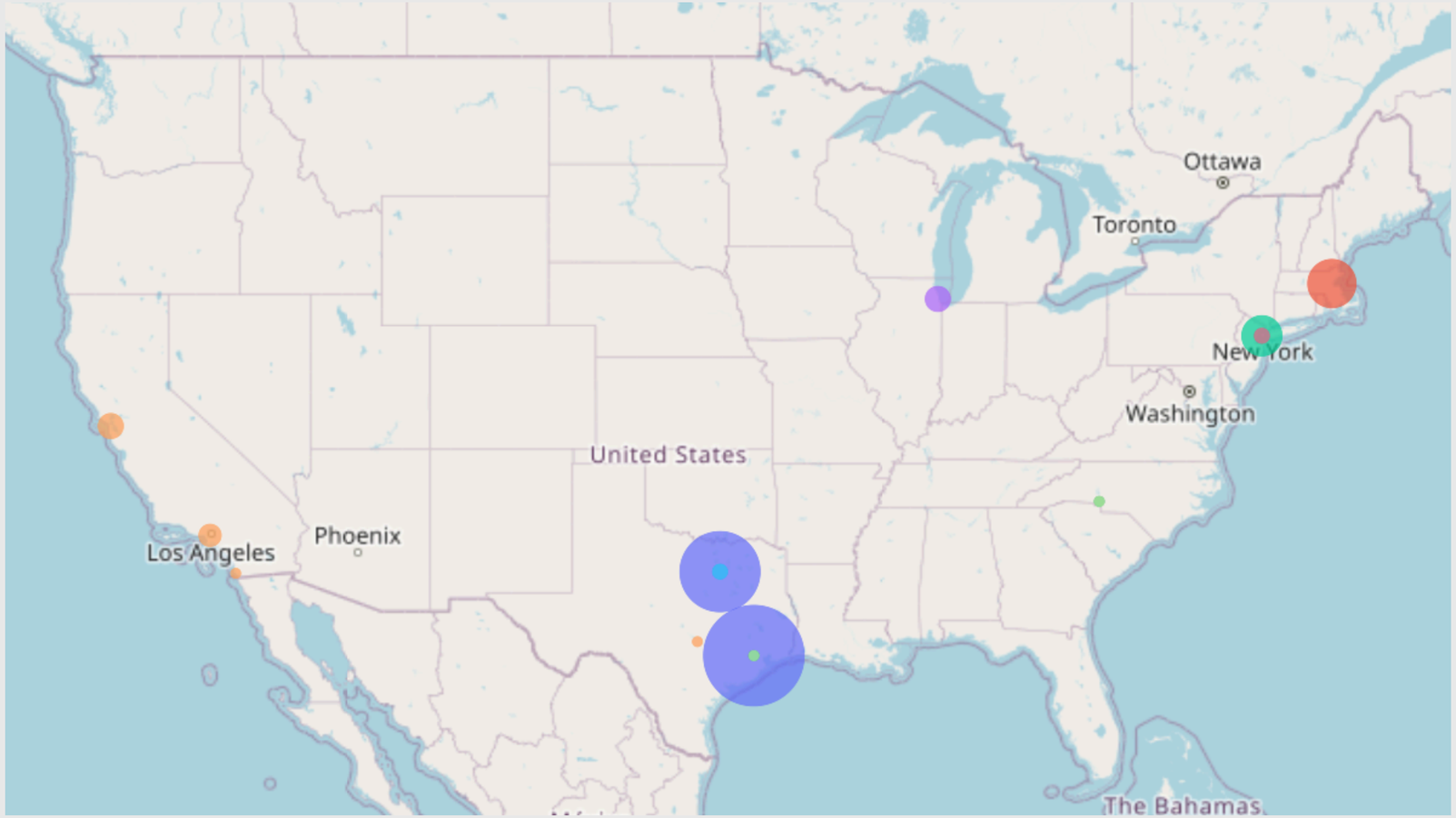
- Focused analysis on major finance hub states

Geospatial Analytics

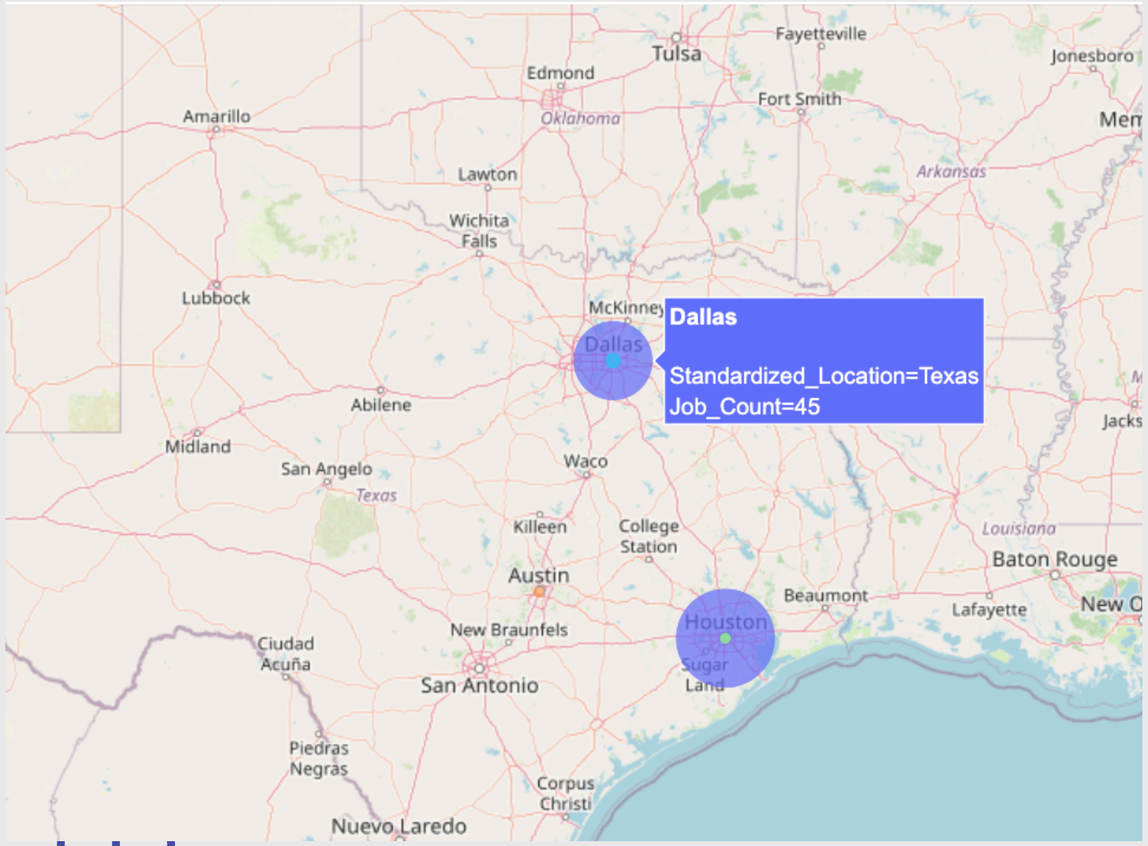


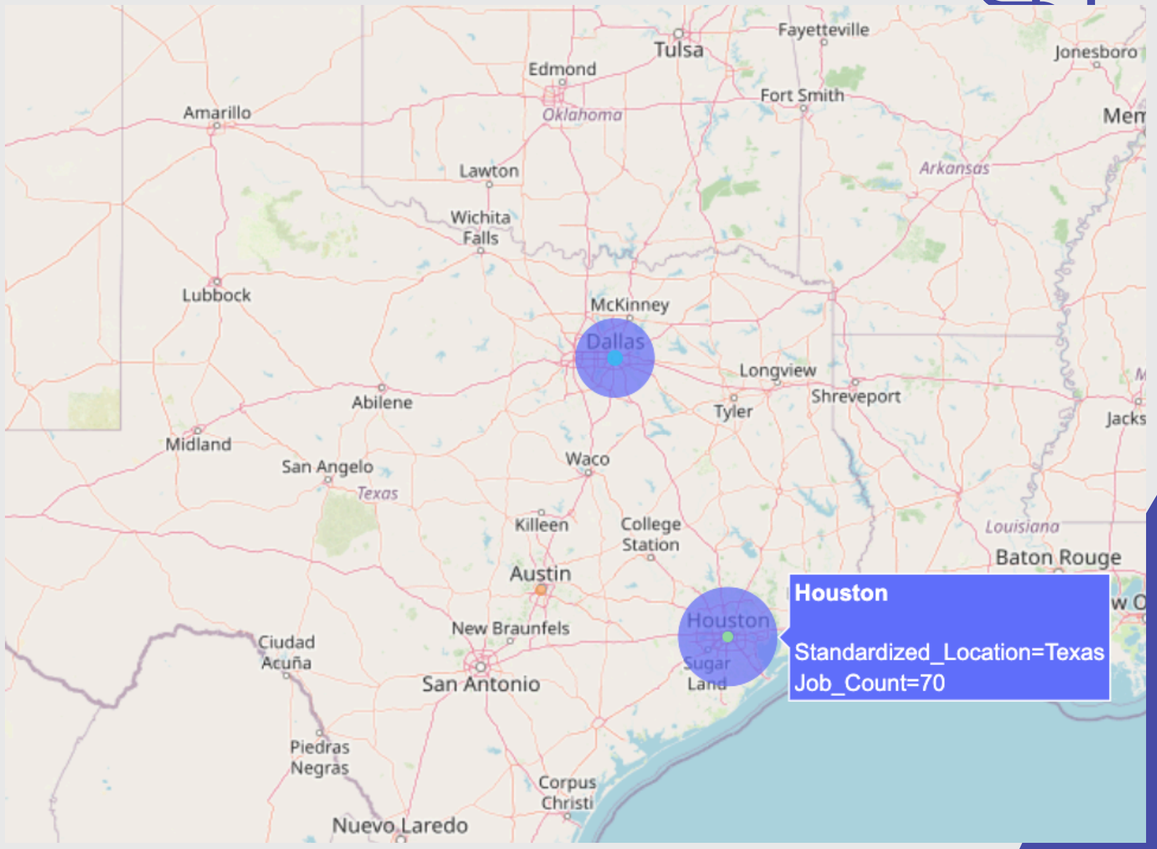
Based on cleaned and merged location data from both profile and tweet text.

City Postings



Texas Focused





Summary and Recommendations

	Metric	Value
0	Total Tweets	604
1	Vocabulary Size	993
2	Remote Job Share (%)	8.51
3	Salary Mentions	0
4	Number of Topics	3
5	Number of Clusters	9
6	Geographic Coverage (States)	8

Thank you